ADHAM ABOULKHEIR

PHD RESEARCHER · EXPLAINABLE AI · AI ENGINEER

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PROFILE

PhD researcher in artificial intelligence at the University of Essex, working on explainable fuzzy classifiers for living neural biocomputers, alongside an AI Engineer post on the UK Data Service's QUADRA-AI project and teaching support across two postgraduate modules. Three years of prior industry experience delivering production machine learning at BT Group and Saudi Motorsport, spanning generative models, retrieval-augmented LLM systems, computer vision and real-time streaming inference. Published at IEEE WCCI FUZZ-IEEE 2026. Particular interest in problems where a model's decision has to be defensible, not merely accurate.

EDUCATION

PhD in Artificial Intelligence
2025 — Present

University of Essex, United Kingdom

- Supervised by Prof. Hani Hagras and Dr. Michael Barros
- Research: explainable AI and fuzzy rule-based classifiers for living neural biocomputers grown on Multi-Electrode Arrays

Industrial MSc in Artificial Intelligence
2023 — 2024

University of Essex, United Kingdom

BSc in Computer & Communication Engineering
2019 — 2024

Alexandria University, Egypt

PUBLICATIONS

A Fuzzy-Based Approach for Interpretable Spike Detection in Living Neural Biocomputers
2026

IEEE WCCI · FUZZ-IEEE 2026 · Maastricht, Netherlands

- Interpretable fuzzy rule-based system for spike detection on Multi-Electrode Arrays
- 97.74% F1-score across six biologically diverse chips using 337 human-readable IF-THEN rules
- ANNIGMA feature selection with genetic-algorithm-optimised rule weights

EXPERIENCE

AI Engineer — Disclosure Protection, Anonymisation & Transcription
Oct 2026 — Present

QUADRA-AI · UK Data Archive / UK Data Service, University of Essex

- Applying AI to automated disclosure protection, anonymisation and transcription across the UK Data Service's research data holdings
- The UK Data Service holds the UK's largest collection of social, economic and population research data

Graduate Laboratory Assistant
Oct 2026 — Present

University of Essex, School of Computer Science and Electronic Engineering

- Teaching support across two postgraduate modules, CE801 and CE889
- Lab demonstrating, student support and assessment

AI Product & Backend Delivery Engineer
Jun 2025 — Oct 2026

ThresholdXpert AI Coach

- Designed and built the athlete performance analysis backend, moving the product from prototype to a stakeholder-ready Phase 3 release
- Fatigue modelling using ATL/CTL training-load metrics and a personalised recommendation engine
- FastAPI endpoints for session upload and readiness scoring

AI Researcher

Oct 2023 — Jan 2025

BT Group

- Built a DCGAN + Beta-VAE + Stable Diffusion pipeline expanding 50 real telecom images into 350,000+ synthetic training images, improving YOLOv8 mAP@0.5 by 9.6 percentage points to 0.943 across ten fault classes
- Evaluated generation quality with FID and Inception Score, and validated utility through a downstream mAP ablation study
- Built an LLM + RAG fault diagnosis system (LangChain, FAISS, FastAPI) achieving 87.3% resolution accuracy and enabling remote resolution without engineer site visits

Associate AI Software Engineer

Nov 2023 — Oct 2024

Saudi Motorsport Company

- Built a real-time ML system processing 13 telemetry channels per lap — throttle, brake, tyre temperatures and G-forces
- Streaming anomaly detection, lap-time prediction and tyre degradation modelling over a Kafka transport layer
- Containerised with Docker and orchestrated with Kubernetes for race-weekend reliability

SELECTED PROJECTS

Biocomputer Explainability Framework

2026

PhD research

- Electrode importance (ANNIGMA and permutation), fuzzy rule extraction, counterfactual explanations, Moran's I spatial autocorrelation and functional hub detection

Agentic AI with LangChain & LangGraph

2025

Open source

- Multi-agent workflow system for autonomous multi-step execution with tool use, memory and human-in-the-loop checkpoints

Generative AI & LLM Pipeline

2025

Open source

- LoRA/QLoRA fine-tuning for Mistral-7B, hybrid TF-IDF RAG, chain-of-thought prompting, faithfulness scoring and BLEU/ROUGE benchmarking — 78% reduction in hallucination against an ungrounded baseline

Interval Type-2 Fuzzy Classifier

2025

PhD research

- IT2 fuzzy classification with GA-optimised interval-valued rule weights for modelling second-order uncertainty

Predictive Maintenance Pipeline

2024

Open source

- Rolling feature engineering with an XGBoost failure predictor at AUC-ROC 0.97 and 48–72 hour lead time, MLflow-ready

Full index of 19 public repositories at github.com/Adham5172001

SKILLS

Research	Explainable AI · Fuzzy rule systems (T1 & IT2) · Genetic algorithms · Neural biocomputing · MEA analysis · Feature selection · Spatial statistics · Statistical methods
Machine learning	Deep learning · Generative AI · LLMs & RAG · Computer vision · NLP · Time series · Reinforcement learning
Engineering	Python · PyTorch · TensorFlow · Scikit-learn · HuggingFace · LangChain · LangGraph · FastAPI · Docker · Kubernetes · Kafka · AWS · MLflow · Git · Linux